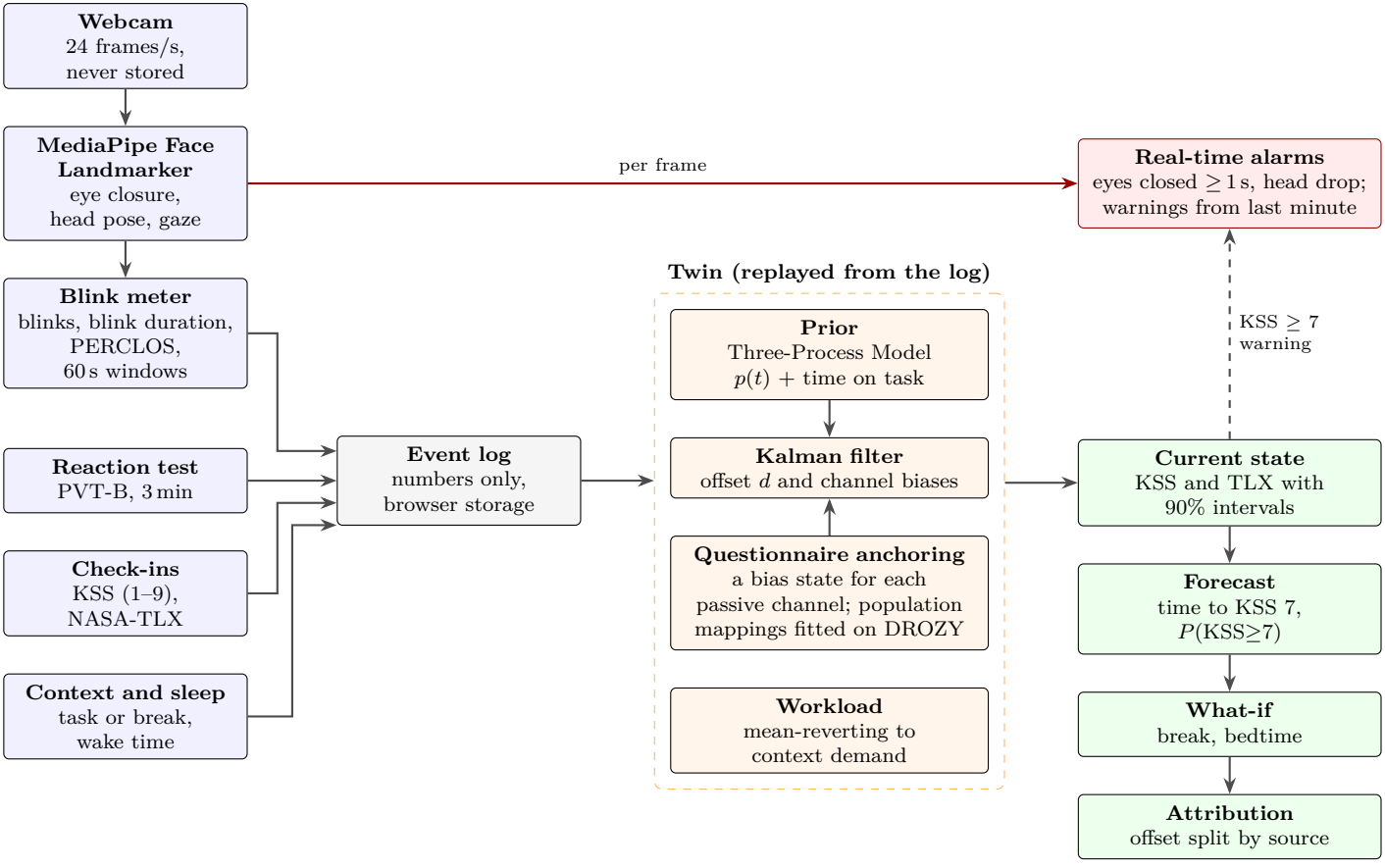

Webcam
24 frames/s, never stored
MediaPipe Face Landmarker
eye closure, head pose, gaze
Blink meter
blinks, blink duration, PERCLOS, 60 s windows
Reaction test
PVT-B, 3 min
Check-ins
KSS (1–9), NASA-TLX
Context and sleep
task or break, wake time
Event log
numbers only, browser storage
per frame
Real-time alarms
eyes closed ≥ 1 s, head drop; warnings from last minute
Twin (replayed from the log)
Prior
Three-Process Model
p(t) + time on task
Kalman filter
offset d and channel biases
Questionnaire anchoring
a bias state for each passive channel; population mappings fitted on DROZY
Workload
mean-reverting to context demand
KSS ≥ 7 warning
Current state
KSS and TLX with 90% intervals
Forecast
time to KSS 7, P(KSS≥7)
What-if
break, bedtime
Attribution
offset split by source